

Daniele Alessi

Contacts

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Personal Website & Portfolio:

<https://eldanee.github.io>

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<https://www.linkedin.com/in/daniele-alessi/>

Education

Master degree in Computer Science Engineering @ Politecnico di Milano
(September 2021 - April 2025)

Erasmus Programme @ IST Universidade de Lisboa
(February 2023 - July 2023)

Bachelor degree in Computer Science Engineering @ Politecnico di Milano
(September 2018 - September 2021)

Experience

Graphics & Rendering Junior Programmer @ Untold Games; Genoa, Italy
(June 2025 - Present)

IT Specialist Intern @ Bending Spoons; Milan, Italy
(March 2019 - September 2020)

Skills

Computer Graphics

- Real time rendering and GPU accelerated ray tracing using **Vulkan** and **GLSL**
- Practical and theoretical knowledge of various culling and rendering techniques
- Experience with Vulkan and GLSL, and basic knowledge of many other graphics API and shading languages.

GPU programming

- GPU accelerated Position Based Dynamics simulation using **Vulkan Compute** for my master thesis
- GPU profiling using RenderDoc, Pix and various platform specific tools.
- Basic knowledge of **CUDA**

Video game Design and Programming

- Designed and programmed a video game in **Unity** for an academic project (Beta version available at <https://polimi-game-collective.itch.io/diagonaliz>)
- Smaller projects for various Game Jams, usually in Godot
- Basic experience with consoles SDKs and tool suites.
- Junior experience working on Unreal Engine 4, adding and adapting features to the engine codebase.

Other

Languages

Italian(native speaker), **English**(proficient in both speaking and writing)

Best Paper Award 2025 @ ASME VARE

Received the ASME Virtual and Augmented Reality Environments Best Paper Award 2025 at the IDETC/CIE 2025, presenting the paper "*Efficient Soft Body Simulation for Real-Time Applications: A GPU-Based XPBD Approach*", Alessi, D, Buttiglione, MD, Lucania, E, Piazzolla, P, Colombo, G, Gribaudo, M.
The abstract can be found [here](#).